

Narrative Synthesis Report: Parameterizing Infectious Disease Models in Maritime Environments

Executive Summary

This report synthesizes evidence from 20 scientific documents and reports to provide parameterization data for cruise ship outbreak modeling, with a focus on 10 major pathogens including SARS-CoV-2, norovirus, influenza, and various enteric bacteria. Modeling infectious disease in a maritime context requires accounting for the "hidden" spectrum of infection, where clinical attack rates reported by systems like the Vessel Sanitation Program (VSP) represent only a subset of total infections (B2). This synthesis addresses the critical need to quantify asymptomatic and subclinical proportions, transmission mechanisms (aerosol, fomite, and person-to-person), and the impact of public health responses on observed data.

The analysis of these 20 papers indicates that asymptomatic proportions vary significantly by pathogen, with estimates for SARS-CoV-2 ranging from 17.9% to 74% (B2), influenza reaching approximately 75% (B1), and norovirus exhibiting high individual heterogeneity in transmission, often driven by "superspreaders" with mild symptoms and long diagnostic delays (B2). Key takeaways include the dominance of aerosol inhalation in SARS-CoV-2 spread (B1), the necessity of high-frequency disinfection and handwashing thresholds (>70%) to mitigate contact transmission (B1), and the utility of viral metagenomics to uncover unidentified etiological agents in clinical samples (B1). This report provides the necessary data to refine model parameters beyond clinical reporting, accounting for biological shedding, environmental persistence, and behavioral factors.

Thematic Findings

Asymptomatic and Subclinical Infection Proportions

A primary challenge in parameterizing maritime outbreak models is the high frequency of asymptomatic or subclinical infections that evade standard surveillance. For SARS-CoV-2, the *Diamond Princess* outbreak provided a unique cohort where high testing rates allowed for the estimation of the asymptomatic proportion at 17.9% (95% CrI: 15.5–20.2%) [2]. However, alternative modeling that accounts for a- and pre-symptomatic states suggests this figure could be as high as 74% (95% PI: 70–78%) [1]. In a repatriated cohort from another cruise ship, 42% of cases were asymptomatic [1]. Clinical analysis of 104 patients from the *Diamond Princess* revealed that 32% remained asymptomatic throughout the entire observation period [1].

For other pathogens, the subclinical fraction is equally significant. Influenza infections on ships are characterized by a high rate of asymptomatic cases, estimated at roughly 75% [1]. In norovirus outbreaks, the asymptomatic carrier state is a recognized driver of transmission, particularly when linked to food handlers who remain symptom-free while actively shedding the virus [2]. Metagenomic analysis of gastroenteritis patients who tested negative for common pathogens revealed a variety of "hidden" viruses, including norovirus GIV and Saffold viruses, which are not routinely screened and may contribute to the subclinical burden [1].

Transmission Dynamics: Aerosol, Fomite, and Contact

Mechanistic modeling of the *Diamond Princess* indicates that aerosol inhalation was likely the dominant contributor to SARS-CoV-2 transmission, accounting for more than 50% of cases [1]. This includes both short-range and long-range aerosol transport, even under conditions of high ventilation and no air recirculation [1]. Environmental sampling supported this, as SARS-CoV-2 RNA was detected on cabin surfaces for up to 17 days post-vacation, specifically on bathroom floors and pillows, though no live virus could be isolated [1].

In contrast, norovirus dynamics on ships are predominantly driven by person-to-person and fomite routes [1]. Reconstructed transmission chains for norovirus reveal that 88% of infections occurred in public spaces, with limited transmission within cabins [1]. Contact infection models suggest that virus distribution in seating areas often follows a "V-shape," where seats near main aisles are cleaner due to high-frequency contact and subsequent "dilution" of the virus by many hands, while seats near walls or in low-traffic areas accumulate higher viral loads [1]. The transfer rate between surfaces is a key factor; a

transfer rate of approximately 0.25 was found to minimize infection risk by balancing viral spread and dilution (1).

Diagnostic Latency and "Superspreading" Events

Parameterizing the timing of the public health response is critical. In a norovirus outbreak analysis, 10% of infected individuals (superspreaders) were responsible for 57% of secondary cases (1). These individuals were characterized by a halved frequency of symptoms and significantly longer diagnostic delays—an average of 83 hours compared to 47 hours for other infectors (1). This suggests that "mild" cases are more dangerous for model outcomes because they remain in the population longer.

For influenza, sudden onset and high attack rates (up to 37%) necessitate rapid intervention (2). On navy vessels, crowded living conditions (e.g., >50 persons per sleeping compartment) increased the risk of infection by 2 to 3 times, highlighting the importance of population density parameters in military and maritime models (2). Genomic analysis on the *USS Theodore Roosevelt* confirmed that even among young, healthy adults, asymptomatic carriage is high (initially 76.9%), making symptom-based screening fallible (1).

Effectiveness of Public Health Interventions

Modeling and empirical data suggest that traditional maritime quarantine measures, such as confining passengers to cabins, have mixed results. While movement restrictions on the *Diamond Princess* reduced the basic reproduction number (R_0) from an initial high of 14.8 to 1.78, an earlier evacuation would have likely resulted in only 76 infections instead of the observed 712 (2).

Specific behavioral parameters are highly sensitive. Contact models show that the turned point for the impact of handwashing and disinfection occurs when both probability and efficiency exceed a 70% threshold (1). Below this, the virus spreads effectively regardless of the intervention (1). In norovirus management, a 72-hour isolation protocol was found to avert 71% of potential cases (2). For respiratory viruses, rapid masking and the elimination of group events are essential, as demonstrated on the *Grand Princess*, where these measures were implemented following the detection of cases from a previous voyage (1).

Consensus & Contradictions

Consensus on Mode of Transmission

There is broad consensus across the literature that person-to-person transmission is the primary driver of outbreaks in semi-closed maritime environments (B1 5). The importance of common-source exposures (e.g., food or water) has decreased over time due to strict adherence to programs like the VSP (B1 2). Specifically, waterborne outbreaks have not been frequently reported since 2002, likely due to improved chlorination and cross-connection controls (B1 1).

Contradictions in Asymptomatic Rates

A significant contradiction exists regarding the true asymptomatic proportion of SARS-CoV-2. Statistical modeling by Emery et al. (B1 1) and Mizumoto et al. (B1 1) returned divergent results for the same outbreak (*Diamond Princess*). Emery et al. estimated a 74% asymptomatic rate (B1 1), while Mizumoto et al. initially estimated 17.9% (B1 1). These discrepancies stem from different assumptions regarding "right-censoring" and whether cases labeled as asymptomatic at the time of testing eventually developed symptoms. This highlights a critical parameterization risk for models: the definition of "asymptomatic" vs. "pre-symptomatic" significantly alters the predicted source of transmission (B1 2).

Agreement on Environmental Resilience

The literature agrees that environmental surfaces play a role in maintaining outbreaks, but likely not as the primary driver compared to direct contact or aerosols. Studies on both norovirus and SARS-CoV-2 emphasize that while viral RNA or antigens persist on surfaces (up to 17 days for SARS-CoV-2 (B1 1) and several days for norovirus (B1 1)), the inability to recover live virus in many samples suggests a rapid loss of infectivity (B1 2).

Pathogen-Specific Contributions

Norovirus and Norwalk Virus

Norovirus attack rates on cruise ships are significantly higher for passengers (weighted average 7%) than for crew (2%) (B1). This is attributed to higher passenger density in public areas and potential immunity/resistance among the younger crew population (B2). Specific risk factors include the consumption of "cold dishes" prepared by bare hands (e.g., potherbs, garlic sprouts) (B1) and contaminated ice (B2). Aircraft carrier data emphasizes that crowding is the dominant risk factor for norovirus transmission (B1).

Influenza and Legionella

Influenza outbreaks on ships cause widespread morbidity, with one study reporting 40 hospitalizations and two deaths among 1,100 passengers (B1). Vaccination of at least 80% of the crew is recommended to prevent these outbreaks (B1). Legionellosis is a persistent risk in maritime water systems; a study of 21 ferries found that 66.7% of water distribution systems were colonized by *Legionella* spp. (B1). Whirlpool spas have been specifically implicated in these outbreaks (B1).

Enteric Pathogens (Salmonella, E. coli, Shigella, Vibrio)

While less common than norovirus, bacterial gastroenteritis remains a threat. *Salmonella* spp. are the most frequent bacterial agents, followed by *E. coli* and *Shigella* (B1). Outbreaks are often linked to shore excursions where passengers consume local food (B1). Metagenomic testing has shown that many "untyped" gastroenteritis cases are actually caused by diverse viruses like astrovirus, sapovirus, and parechoviruses (B1).

Research Gaps & Future Directions

1. **Longitudinal Asymptomatic Follow-up:** There is a lack of high-quality data tracking the full clinical course of asymptomatic individuals to distinguish them from pre-symptomatic individuals (B2).
2. **Viral Shedding in Asymptomatic Cases:** Further experiments are needed to quantify the infectivity of asymptomatic carriers vs. symptomatic cases, as current models are forced to return uniform priors for this parameter (B1).

3. **Metagenomic Surveillance:** Routine clinical screening is often too narrow. Incorporating metagenomics could identify why models fail to fit observed data for "unknown" etiological agents  1.
4. **Ventilation Design:** While aerosol transmission is identified as dominant, the specific impact of different HVAC designs on maritime vessels remains under-studied  2.

Limitations of This Review

This synthesis is based on an automated analysis of a limited corpus of 20 documents. It focuses heavily on the 2020-2021 period for SARS-CoV-2 data, which may not reflect the impact of subsequent variants. Furthermore, access to full ship manifests and detailed clinical records for disembarked passengers is often restricted, potentially biasing estimates of attack rates and asymptomatic proportions  2. The systematic reviews included may be subject to publication bias, favoring reports of successful investigations or larger outbreaks  1.

Analyzed Sources

- [1] Epidemiology of Norwalk virus during an outbreak of acute gastroenteritis aboard a US aircraft carrier
- [2] Passenger Behaviors Associated with Norovirus Infection on Board a Cruise Ship—Alaska, May to June 2004
- [3] Environmental sampling for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) during a coronavirus disease (COVID-19) outbreak aboard a commercial cruise ship
- [4] First epidemic of hantavirus on cruise ships: from passenger surveillance to ship ecology
- [5] A large outbreak of influenza A and B on a cruise ship causing widespread morbidity

[6] Outbreak of acute gastroenteritis caused by norovirus genogroup II attributed to contaminated cold dishes on a cruise ship in Chongqing, China, 2017

[7] Asymptomatic COVID-19; We don't know what we don't know

[8] Systematic literature review and meta-analysis on preventing and controlling norovirus outbreaks on cruise ships, 1990 to 2020: calling for behaviour change strategies of travellers

[9] Contact infection of infectious disease onboard a cruise ship

[10] State of the art: public health and passenger ships

[11] Emergence of new norovirus variants on spring cruise ships and prediction of winter epidemics

[12] Unveiling viruses associated with gastroenteritis using a metagenomics approach

[13] Systematic review on outbreaks of SARS-CoV-2 on cruise, navy and cargo ships

[14] Epidemiology of COVID-19 outbreak on cruise ship quarantined at Yokohama, Japan, February 2020

[15] Mitigating norovirus spread on cruise ships: A model-based assessment of diagnostic timing and isolation

[16] The clinical characteristics of COVID-19: a retrospective analysis of 104 patients from the outbreak on board the Diamond Princess cruise ship in Japan

[17] Symptoms and laboratory manifestations of mild COVID-19 in a repatriated cruise ship cohort

[18] Clinical characteristics of COVID-19 in 104 people with SARS-CoV-2 infection on the Diamond Princess cruise ship: a retrospective analysis

[19] Using influenza surveillance networks to estimate state-specific case detection rates and forecast SARS-CoV-2 spread in the United States

[20] Mechanistic transmission modeling of COVID-19 on the Diamond Princess cruise ship demonstrates the importance of aerosol transmission

[21] Estimating the asymptomatic proportion of coronavirus disease 2019 (COVID-19) cases on board the Diamond Princess cruise ship, Yokohama, Japan, 2020

[22] The contribution of asymptomatic SARS-CoV-2 infections to transmission on the Diamond Princess cruise ship

[23] Norovirus outbreak in a cruise ship sailing around the British Isles: investigation and multi-agency management of an international outbreak

[24] Measles, rubella, and varicella among the crew of a cruise ship sailing from Florida, United States, 2006